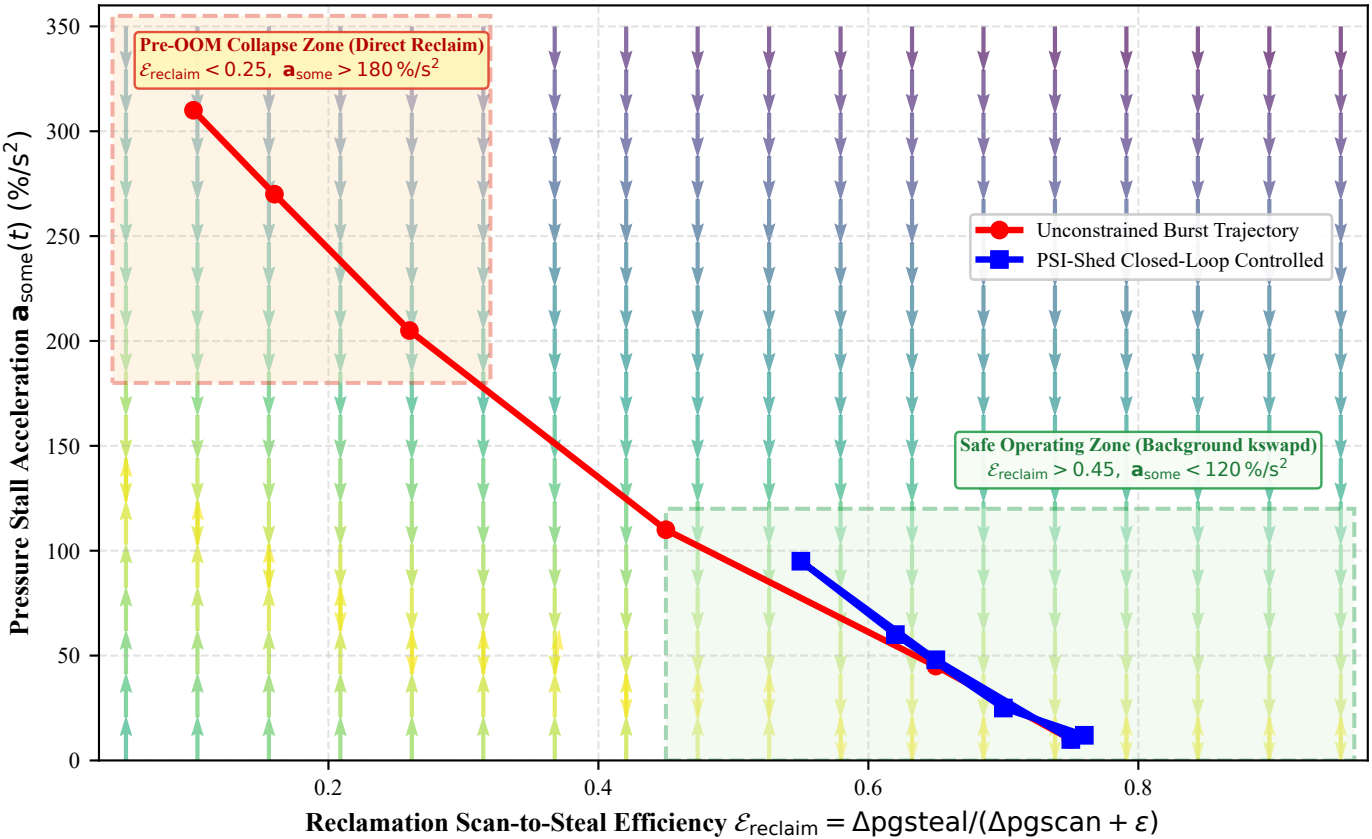

2D Dynamical Phase Plane of Kernel Pressure Stall Kinetics
Pre-OOM Collapse Zone (Direct Reclaim)
$\mathcal{E}_{\text{reclaim}} < 0.25,\ \mathbf{a}_{\text{some}} > 180\ \%/s^2$
Safe Operating Zone (Background kswapd)
$\mathcal{E}_{\text{reclaim}} > 0.45,\ \mathbf{a}_{\text{some}} < 120\ \%/s^2$
Unconstrained Burst Trajectory
PSI-Shed Closed-Loop Controlled
Pressure Stall Acceleration $\mathbf{a}_{\text{some}}(t)$ (%/s^2)
Reclamation Scan-to-Steal Efficiency $\mathcal{E}_{\text{reclaim}} = \Delta\text{pgsteal}/(\Delta\text{pgscan} + \varepsilon)$
0
50
100
150
200
250
300
350
0.2
0.4
0.6
0.8